Lecture II: Explore & Exploit

Programming: Everyday Decision-Making Algorithms

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Today in one sentence

Whether to try something new or stick with what you know depends mostly on how much time you have left.

Why it matters

You have 45 minutes for lunch: the new restaurant or your reliable favorite? Companies face the same choice every day: a new supplier or the proven one, a new marketing channel or the one that works, a new film or another sequel.

Roadmap

1. The dilemma: try something new or stick with what works
2. Time decides: why the end of the interval changes everything
3. Smart strategies: from rules of thumb to the Gittins index

The dilemma: try something new or stick with what works

You have 45 minutes for lunch. Do you try the new restaurant or go to your reliable favorite?

Quick Poll: Raise your hand, who would try the new restaurant?

Photo by Jillian Amatt on Unsplash

Explore gathers information, exploit uses it

Explore

- **Definition:** Gathering new information
- **Risk:** Unknown outcomes
- **Example:** Trying that new sushi place

Exploit

- **Definition:** Using known information
- **Risk:** Missing better options
- **Example:** Your go-to pizza joint

> ! Important
>
> In any given moment you can only do **one of the two**.

Examples from Your Life

Question: **Where do you face explore vs. exploit decisions?**

Explore or exploit: three examples

Medical

Explore: Test a new drug on patients.

Exploit: Use the proven treatment.

Tech

Explore: Test a new website design.

Exploit: Keep the current design.

Dating

Explore: Meet someone new.

Exploit: See someone you already know.

Question: **What is the optimal balance?**

What if you only exploit?

Question: What happens then?

- Exploitation is not always the best strategy
- Especially with **very limited information**
- If you **stop exploring**, you **miss better options**

What if you only explore?

Question: What happens then?

- Exploration is not always the best strategy
- If you **constantly explore**, you **never enjoy the results**
- Imagine eating each meal, hearing each song, meeting each person **only once**

Computer scientists call it the multi-armed bandit

- Computer scientists formalized the explore-exploit trade-off
- It is known as the multi-armed bandit problem
- Question: What is a one-armed bandit?

One Armed Bandit

Photo by Kabir M on Unsplash

Three slot machines, 20 pulls: which do you play?

Imagine: You walk into a casino with **3 slot machines**

- Each machine has a **different** (unknown) win rate
- You have **$100** and **20 pulls** total
- **Goal:** Maximize your winnings
- **Explore:** try different machines. **Exploit:** play the best one so far

Every pull costs money and gives information

- Each pull tells us more about a machine's chance of winning
- But each pull also **costs** money

Question: **How do you decide which machine to play next?**

Interactive Casino Challenge

You've been playing for a while. Here's what you know:

Machine	Pulls	Wins	Win Rate[1]	Your Choice?
A	15	9	60%	
B	2	1	50%	
C	0	0	???	

Class Vote: A (known performer), B (limited data) or C (complete unknown)?

Follow-up: What if you only had **2 pulls** left? Or **100**?

Machine A looks better, but both estimates are shaky

- Expected value = average wins per pull
- E(A) = 9/15 = 0.6 and E(B) = 1/2 = 0.5
- But 2 and even 15 pulls are **few observations**

[1]The win rate is the share of pulls that paid out: wins divided by pulls. Machine A: 9 / 15 = 60%.

The best choice now is not always the best choice overall

- The bandit problem stands for many real-world decisions
- The best choice **now** can differ from the best choice **in the long run**
- Which lever to pull next depends on one more thing

Question: Any idea what it is?

Time decides: why the end of the interval changes everything

Roadmap

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How long will you stay in the casino?

- Question: Why does this matter?
- Question: How does it change our choice between machine A and B?

> "I'm more likely to try a new restaurant when I move to a city than when I'm leaving it" (Chris Stucchio)

Long interval: explore. Short interval: exploit.

- The time you plan to stay in the casino is **the interval**
- **Long interval:** explore more, you have time to use what you learn
- **Short interval:** exploit what you already know

Explore while you still have time to use what you learn

> "I moved to Pune, India, and I just [...] eat everywhere that didn't look like it was gonna kill me" (Chris Stucchio)

Exploration

Photo by Colin Maynard on Unsplash

Exploit when you are ready to cash in

> "And as I was leaving the city I went back to all my old favorites, rather than trying out new stuff [...]. Even if I find a slightly better place, I'm only going to go there once or twice, so why take the risk?" (Chris Stucchio)

Exploitation

Photo by Cristina Gottardi on Unsplash

Hollywood exploits: 9 of the top 10 films are sequels

ℹ Sequels among the 10 highest-grossing films

Year	Number of Sequels
1981	2
1991	3
2001	5
2011	8
2024	9[2]

Question: **What explains this trend, and what does it say about Hollywood's interval?**

Sequels...

Photo by Universal Pictures on Wikipedia

Studios behave as if their interval is ending

- A new movie is risky, but can create a new fan base (**explore**)
- A sequel has a guaranteed fan base, a sure thing (**exploit**)
- Studios may believe they are near the **end of their interval**
- So they pull the best arms before the casino turns them out

Anecdotes are not a strategy

- Anecdotes help us understand the trade-off, but they are no solution
- An exact algorithm for the trade-off is **very hard** to find
- On the way, there were many interesting approaches...

Smart strategies: from rules of thumb to the Gittins index

Roadmap

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[2] Box Office Mojo, top 10 by US domestic gross 2024 (boxofficemojo.com/year/2024). Counted as sequels: all except *Wicked*; *Twisters* counted as a sequel.

3. Smart strategies: from rules of thumb to the Gittins index

Rule of thumb: win-stay, lose-shift[3]

- Question: What do you think this strategy does?
- **Won?** Stay with the same arm. **Lost?** Switch to another arm.
- Simple, and proven better than choosing an arm at random

Win-stay makes sense, lose-shift is too hasty

Win-stay

- If you win, pull the same arm again
- The arm has just become more attractive

Lose-shift

- Switching after every loss is a rash move
- Your favorite restaurant, tenth visit: always great, today disappointing
- Should you never go back?

Easy answers come with problems

- Win-stay lose-shift **penalizes losses too much**
- It ignores **the interval**
- But it was a good start for better approaches

A fair estimate: Laplace's rule of succession

- First we need a fair estimate of each machine
- The plain win rate is extreme after few pulls: 1 win out of 1 pull looks like 100%, 0 out of 0 is undefined
- Laplace (18th century) suggested a fairer estimate: after w wins and l losses, the chance of a win is about

$$\mathbb{E}[\text{win}] = \frac{w+1}{w+l+2}$$

- In words: add one imagined win and one imagined loss, then take the share of wins

Try it: machines A, B and C by Laplace

$$\mathbb{E}[\text{win}] = \frac{w+1}{w+l+2}$$

Question: What is E[B], E[A] and E[C]? Which machine looks better by this estimate?

- E[B] = (1 + 1) / (2 + 2) = 0.50

[3]For more details see Robbins, H. (1952) 'Some aspects of the sequential design of experiments', Bulletin of the American Mathematical Society, 58.

- E[A] = (9 + 1) / (15 + 2) = 10/17 ≈ 0.59
- E[C] = (0 + 1) / (0 + 2) = 0.50, even though C was never played

Laplace gives unknown options a fair chance and avoids estimates of 0% or 100% after a few pulls.

Laplace estimates for wins and losses

Losses/Wins	1	2	3	4	5	6	7	8	9	10
1	0.50	0.60	0.67	0.71	0.75	0.78	0.80	0.82	0.83	0.85
2	0.40	0.50	0.57	0.63	0.67	0.70	0.73	0.75	0.77	0.79
3	0.33	0.43	0.50	0.56	0.60	0.64	0.67	0.69	0.71	0.73
4	0.29	0.38	0.44	0.50	0.55	0.58	0.62	0.64	0.67	0.69
5	0.25	0.33	0.40	0.45	0.50	0.54	0.57	0.60	0.63	0.65
6	0.22	0.30	0.36	0.42	0.46	0.50	0.53	0.56	0.59	0.61
7	0.20	0.27	0.33	0.38	0.43	0.47	0.50	0.53	0.56	0.58
8	0.18	0.25	0.31	0.36	0.40	0.44	0.47	0.50	0.53	0.55
9	0.17	0.23	0.29	0.33	0.38	0.41	0.44	0.47	0.50	0.52
10	0.15	0.21	0.27	0.31	0.35	0.39	0.42	0.45	0.48	0.50

Bellman: calculate backwards from the last pull

- Richard Bellman found the exact solution for a **known interval**
- Start at the **final pull**: with no future left, simply exploit
- Then calculate **backwards** the best choice in every earlier situation

Bellman needs to know how long you stay

- The calculation grows fast with many arms and a long interval
- Worse: we usually **don't know** how long we stay in the casino
- So the bandit problem seemed to have no practical solution

Gittins: a drug-trial problem at Unilever[4]

- In the 1970s, Unilever had several **new compounds** that might work as drugs
- One test = one patient. **Improvement = win, no improvement = loss**
- Testing costs **time and money**, and nobody knows in advance which compound is best
- Unilever asked the mathematician John Gittins: **which compound should we test next, and when should we stop testing?**

[4] For more details see Gittins, J. (1979) 'Bandit Processes and Dynamic Allocation Indices', Journal of the Royal Statistical Society. Series B (Methodological).

- Gittins saw that this is **exactly our casino**: each compound is a slot machine with an unknown win rate

Gittins discounts the future

- His answer: give every machine a single number, computed only from its own wins and losses, and always play the machine with the highest number. That number is the **Gittins index**.
- Earlier solutions needed a **known interval**
- Gittins instead counts future wins less than immediate ones (**discount factor**), like future cash flows
- **Money:** Money today can earn interest
- **Everyday life:** Tonight's dinner beats the same dinner next week
- So any interval length works, even an endless one

Back to machine A and B

- The Gittins index works for every multi-armed bandit problem
- Let's consider our two machines one last time:
  - Machine A: 15 pulls, 9 wins, 6 losses
  - Machine B: 2 pulls, 1 win, 1 loss

The Gittins index as a lookup table

Gittins index for different numbers of wins and losses (discount factor 0.9[5]).

Losses/Wins	0	1	2	3	4	5	6	7	8	9
0	.7029	.8001	.8452	.8723	.8905	.9039	.9141	.9221	.9287	.9342
1	.5001	.6346	.7072	.7539	.7869	.8115	.8307	.8461	.8588	.8695
2	.3796	.5163	.6010	.6579	.6996	.7318	.7573	.7782	.7956	.8103
3	.3021	.4342	.5184	.5809	.6276	.6642	.6940	.7187	.7396	.7573
4	.2488	.3720	.4561	.5179	.5676	.6071	.6395	.6666	.6899	.7101
5	.2103	.3245	.4058	.4677	.5168	.5581	.5923	.6212	.6461	.6677
6	.1815	.2871	.3647	.4257	.4748	.5156	.5510	.5811	.6071	.6300
7	.1591	.2569	.3308	.3900	.4387	.4795	.5144	.5454	.5723	.5960
8	.1413	.2323	.3025	.3595	.4073	.4479	.4828	.5134	.5409	.5652
9	.1269	.2116	.2784	.3332	.3799	.4200	.4548	.4853	.5125	.5373

[5]The discount factor says how much a win in the next round is worth compared to a win now. 0.9 roughly means we expect about ten more pulls. The longer you expect to stay, the closer the factor to 1, and the more exploring pays off.

Machine A or B by the Gittins index?

- Machine B (0.6346) beats machine A (0.6300)
- Clear **win-stay** pattern, relaxed **lose-shift** pattern
- At (0,0) we see the **exploration bonus**

Gittins Index: the math behind the table

$$G_i(s_i, f_i) := \sup_{\tau \geq 1} \frac{\mathbb{E}\left[\sum_{t=0}^{\tau-1} \beta^t \cdot r_i^t \;\middle|\; s_i, f_i\right]}{\mathbb{E}\left[\sum_{t=0}^{\tau-1} \beta^t\right]}$$

In words: imagine a second machine that pays out at a fixed, known rate. The index is the rate that machine would need, so that you'd be indifferent between the two. The more uncertain a machine, the higher its index, because it might turn out great.

> ! Important
>
> **Too complex to calculate by hand**, and you don't need to: just use the lookup table!

Higher index, better choice

- **Core idea:** "What's the best possible average reward from this option?"
- **Higher index = better choice**
- Less-tested options get an **exploration bonus**, that's why machine B wins
- The index balances exploration and exploitation automatically

Where bandits run today

- **Online advertising:** which of thousands of ads to show to this visitor, learning from every click
- **Website testing:** two versions of a page, and the better one gets more visitors while the test is still running
- **E-commerce and A/B testing:** shifting traffic to the better version while the test is still running
- **Clinical trials:** adaptive trials assign more patients to the treatment that works best so far

The same trade-off as in our casino, decided millions of times a day.

Which rule when?

- Both outcomes known: choose the higher expected value
- E(explore) unknown, interval known: **Bellman** (backward calculation)
- E(explore) unknown, interval unknown: **Gittins index**

What we covered

The dilemma: try something new or stick with what works

Every choice between the new and the known trades information against reward.

Time decides: why the end of the interval changes everything

The more time is left, the more exploring pays off; near the end, exploit what you know.

Smart strategies: from rules of thumb to the Gittins index

Simple rules like win-stay, lose-shift help, but better strategies weigh both what you know and how uncertain you still are.

Why it matters

You have 45 minutes for lunch: the new restaurant or your reliable favorite? Companies face the same choice every day: a new supplier or the proven one, a new marketing channel or the one that works, a new film or another sequel.

Explore early.

Exploit late.

Literature

Interesting literature to start

- Christian, B., & Griffiths, T. (2016). Algorithms to live by: the computer science of human decisions. First international edition. New York, Henry Holt and Company.[6]

Books on Programming

- Downey, A. B. (2024). Think Python: How to think like a computer scientist (Third edition). O'Reilly. Free online
- Elter, S. (2021). Schrödinger programmiert Python: Das etwas andere Fachbuch (1. Auflage). Rheinwerk Verlag.

> **Note**
>
> Think Python is free online and a great start. Schrödinger programmiert Python is a playful alternative in German.

[6] The main inspiration for this lecture. We have read it and discussed it in depth, always wanting to translate it into a course.

More Literature

For more interesting literature, take a look at the literature list of this course.